

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M.Tech Examination (CE)

May 2013

CE703 : Algorithms & Computational Complexity

Date: 09.05.2013, Thursday

Time: 10:00 a.m to 1:00 p.m

Maximum Marks:70

Instructions:

1. Attempt all questions.
2. Answers both sections in separate answer books.
3. Figure to the right indicates full marks.
4. Assume suitable data, if necessary and mention it.

SECTION-I

Q-1

- (a) Explain Greedy-choice property and Optimal substructure property by an example. [04]
- (b) Define Best case, Worst Case and Average Case Time Complexity of an algorithm with the help of a suitable example. [03]

Q-2

- (a) Sort the given numbers using merge sort: 85,24,63,45,17,31,96,50,27,7,90 Also determine asymptotic cost of Divide, Conquer and Combine using recurrence tree method. [07]
- (b) What is Spanning tree? Give the Generic algorithm to find Spanning Tree for a given graph. [07]
List down similarities and differences between Kruskal and Prim's algorithm.

OR

Q-2

- (a) State Master Theorem and Solve following recurrence using Master Theorem. [07]
 1. $T(n) = 9T(n/2) + n$
 2. $T(n) = T(2n/3) + 1$
- (b) Give the Difference between Branch & Bound and Backtracking approach. Solve the following task assignment problem instance for minimization using Branch & Bound Method.

	1	2	3
a	4	7	3
b	2	6	1
c	3	9	4

Q-3 Attempt Any Two.**[14]**

- (a) What is the use of Recurrence Tree? Solve the given recurrence using recurrence Tree method. $T(n)=3T(n/4)+n$
- (b) Sort the following list in increasing order of numbers using quick sort. Choose first element as a pivot element.
9, 94, 45, 47, 28, 98, 65, 42, 78, 4, 11, 88, 6
- (c) Solve the following recurrence exactly.

$$t_n = \begin{cases} n & \text{if } n=0,1 \text{ or } 2 \\ 5t_{n-1} - 8t_{n-2} + 4t_{n-3} & \text{otherwise} \end{cases}$$

SECTION-II**Q-4**

- (a) What is dynamic programming? How it is differ from divide-and-conquer? **[03]**
- (b) What is Asymptotic Notation? Explain O , Ω , θ notation with a suitable example. **[04]**

Q-5

- (a) Discuss Dijkstra's algorithm to solve the single source shortest paths on a weighted, directed graph with a example. **[07]**
- (b) Consider $n=3$, $(w_1, w_2, w_3) = (2, 3, 3)$ and $(p_1, p_2, p_3) = (1, 2, 4)$ and $m=6$ find the optimal solution using dynamic programming for 0/1 knapsack problem using given data. **[07]**

OR**Q-5**

- (a) Compare the naïve matching, Rabin Karp and Finite Automata string matching algorithms with respect to preprocessing and matching time. **[07]**
- (b) Given a finite set of distinct coin types, say 50, 20, 10, 5, 2, 1 pcs, and an integer amount C . Each type is available in unlimited quantity. Write an algorithm to find the exact change with minimum no. of coins. What strategy will you use? What is the complexity of algorithm? **[07]**

Q-6 Attempt Any Two.**[14]**

- (a) For the following four matrices find the order of parenthesization for the optimal chain multiplication.
 $A_1 = 15 \times 5$, $A_2 = 5 \times 10$, $A_3 = 10 \times 20$, $A_4 = 20 \times 25$.
- (b) Determine the Logest Common Subsequence of the given two string : $S_1 : ALGORITHM$
 $S_2 : LOGERITHM$. Show all the steps
- (c) Generate at least 2-solutions for 4-Queen's problem using backtracking. Draw a pruned state space tree for the problem.
